



CENTRAL UNIVERSITY OF SOUTH BIHAR

CUSB1902332028

Department of Computer Science

End Semester Examination, Dec 2025

Subject: Big Data Analytics Course Code: CAI91DC01904 Semester : Third

Programme : M. Sc. (AI)

Session: 2024-2026

Time: 3 Hours

M. M.: 70

Course Credit : 4

Enrollment Number (to be filled by Student):

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis.

Section – A

Answer All the questions in not more than 50 words. Each question carries 1 Points.
(1 x 10 = 10 Points)

1. Write down the Gartner's definition of Big Data.
2. What are the phases of Data Evolution by Chen et al.?
3. List the four big data challenges.
4. Explain the AWS S3.
5. What is the use of ClickHouse tool?
6. What are the motivation behinds the HDFS.
7. What is Zookeeper? List the benefits of it.
8. What are the disadvantages of Databricks?
9. What is the difference between replication and sharding?
10. Explain the importance of Big data visualization.

Section – B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.
(6 x 5 = 30 Points)

1. Discuss Big Data in terms of five dimensions, volume, velocity, variety, veracity and Value.
2. What is CAP theorem? We have 1000 different applications which access a NoSql database for lookup. The lookup tables are static and its content get refreshed less frequently. If my application can tolerate working with stale data as well, which category of Nosql database should opt for (CA, CP, PA according to CAP theorem) and why?

[04+02 =06]

3. In which category (structured, semi structured or unstructured) you will place web pages, power point presentations, word documents, e-mail and CCTV footage? Also justify your answer.
4. Write short notes on any two
- a) Big Table
 - b) MongoDB
 - c) Apache Cassandra.
- [03+03 =06]
5. Explain the Apache Spark and Snowflake Analytics Tools.
6. Differentiate between
- (a) Zoho Analytics and Google BigQuery Analytics
 - (b) Tableau and Power BI data visualization

Section – C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points.
(15 x 2 = 30 Points)

- 1.
- a) Explain the big data applications in our educational system and health care system.
 - b) Illustrate the difference types of Big Data Analytics using example/application.
- [5+10=15]
- 2.
- a) Explain in brief the different types of Database Sharding Techniques.
 - b) Explain the different types of (key-value, document, column and graph) NoSql data stores using suitable example.
- [5+10=15]
- 3.
- a) Why we need the Hadoop ecosystem?
 - b) Explain the architecture of Hadoop ecosystem and working of Hadoop Distributed File System (Master-Slave) in detail.
 - c) Illustrate the Hadoop MapReduce concept using suitable example.
- [2+8+5=15]